

```
typewriter-mimicker
=====
```

Plain text to typewriter-style PDF.

The project has two deliberately separate stages.

1. formatter

Haskell. Parses plain text conventions, wraps prose, preserves preformatted blocks, paginates a fixed-width page, and emits a deterministic glyph JSON model.

2. renderer

Python. Reads the glyph model, rasterizes each glyph with light typewriter jitter and ink variation, adds a paper background, and writes a raster PDF. It also adds subtle line-level vertical drift to mimic small carriage return and paper advance variation.

The default renderer font is the bundled Special Elite typewriter face. It is licensed under the Apache License 2.0. See:

```
assets/fonts/SpecialElite-Apache-2.0.txt
```

Usage

Enter the development shell:

```
nix develop
```

Render in one command:

```
nix run . -- examples/sample.txt .artifacts/sample.pdf
```

Render with optional aged paper:

```
nix run . -- examples/sample.txt .artifacts/sample.pdf --paper aged
```

Options

The one-command Nix app accepts renderer options after the output path:

```
nix run . -- INPUT.txt OUTPUT.pdf --dpi 150 --paper aged
```

Formatter options:

```
--cols N
```

Number of fixed-width text columns. Default: 72.

```
--rows N
```

Number of text rows per page. Default: 58.

```
--paper NAME
```

Logical paper name in the JSON page model. Default: letter.

--seed N
Base seed for deterministic glyph effects. Default: 1.

Renderer options:

--dpi N
Output raster resolution. Default: 300.

--preset NAME
Visual effect preset. Default: light-typewriter.
The clean preset disables jitter, line drift, ink bleed,
and missing ink.

--paper clean|aged
Paper treatment. Default: clean.

--font FONT.ttf
Override the bundled Special Elite font.

Generate glyph JSON:

```
mkdir -p .artifacts
stack --stack-yaml formatter/stack.yaml run -- examples/sample.txt \
  > .artifacts/pages.json
```

Render a PDF:

```
uv --project renderer run python renderer/main.py \
  .artifacts/pages.json \
  --output .artifacts/sample.pdf \
  --dpi 150
```

Run checks:

```
stack --stack-yaml formatter/stack.yaml test

uv --project renderer run pytest
```